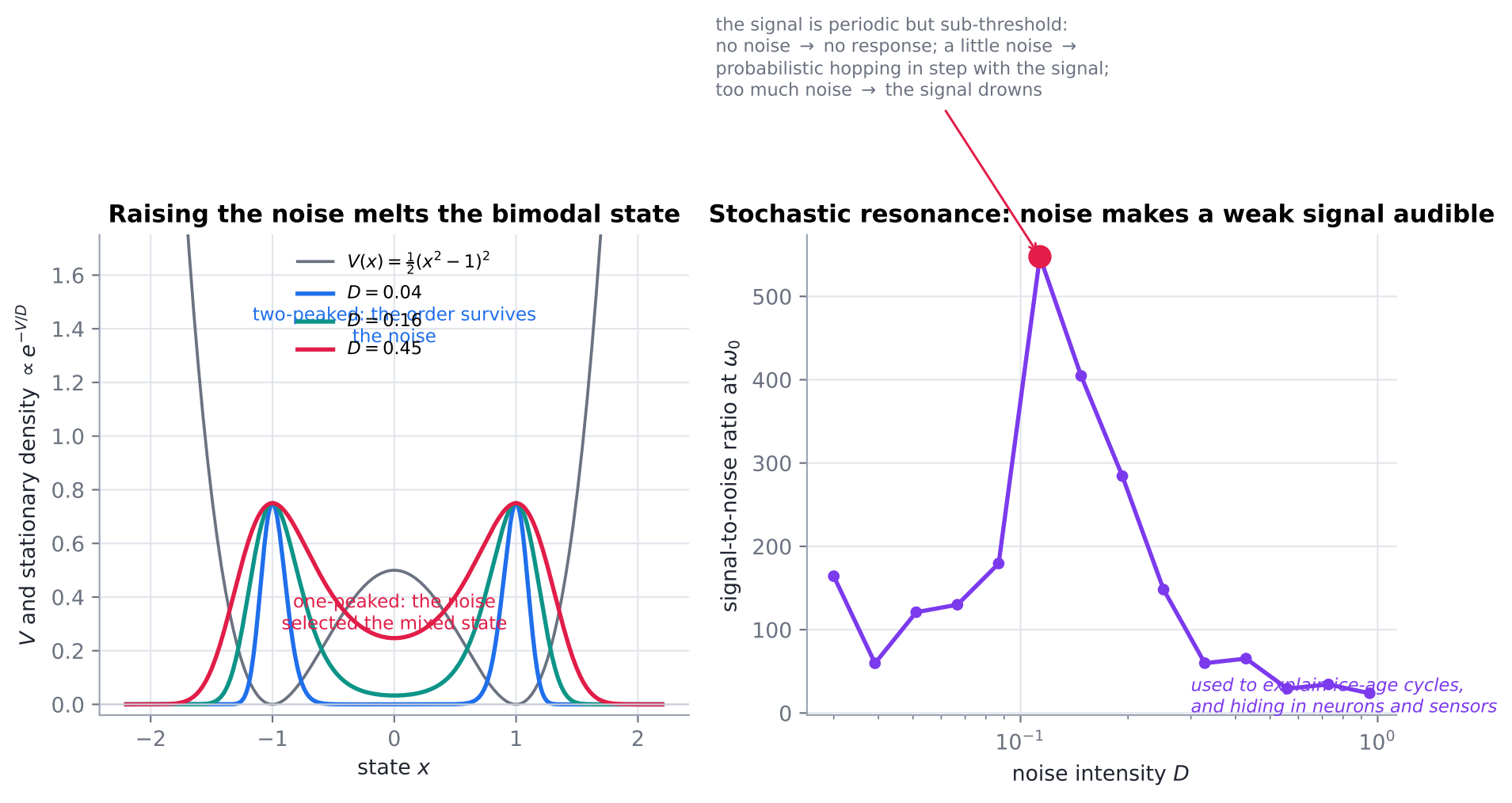
Raising the noise melts the bimodal state
$V(x) = \frac{1}{2}(x^2-1)^2$
$D = 0.04$
$D = 0.16$
$D = 0.45$
two-peaked: the order survives the noise
one-peaked: the noise selected the mixed state
V and stationary density $\propto e^{-V/D}$
state x
Stochastic resonance: noise makes a weak signal audible
the signal is periodic but sub-threshold:
no noise → no response; a little noise →
probabilistic hopping in step with the signal;
too much noise → the signal drowns
signal-to-noise ratio at ω_0
noise intensity D
used to explain ice-age cycles,
and hiding in neurons and sensors